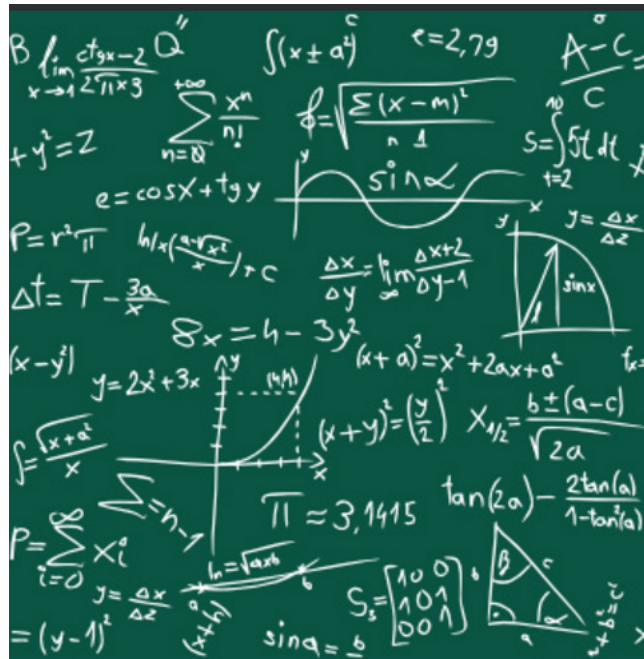


Principle of mathematical induction



- **Author:** Fabio Furini.
- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani and Sandro Salsa, Zanichelli.

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1 The principle of mathematical induction

- We now present a proof technique called proof by induction. This procedure can be applied to theorems with the following structure:

“ for every $n \in \mathbb{N}$, $n \geq n_0$, property $p(n)$ holds ”

- The number n_0 is the smallest integer for which we want the property to be true; if $n_0 = 0$ the theorem simply states that the property is true for every $n \in \mathbb{N}$.

A proof by induction consists of the following two steps:

1. We prove that $p(n)$ is true for $n = n_0$ (**base case of the induction**).
2. We prove that, if n is an arbitrary natural number $\geq n_0$, the fact that $p(n)$ is true implies that $p(n + 1)$ is true (**inductive step**).

We can then conclude that for every $n \geq n_0$, $p(n)$ is true.

- The validity of this proof technique, *intuitively*, is based on the following fact:
 1. By point 1, we know that $p(n_0)$ is true. Suppose for example $n_0 = 1$: we therefore know that $p(1)$ is true (this must be proved explicitly).
 2. By point 2, since $p(1)$ is true, $p(2)$ will be true: indeed, we have proved that for any n , if $p(n)$ is true then $p(n + 1)$ is also true. But then, since $p(2)$ is true, $p(3)$ will be true; but then $p(4)$ is true, ... and so on, hence $p(n)$ is true for every $n \geq 1$.
- In practice, the proof consists of two phases.
 1. Prove $p(n_0)$ directly;
 2. Assume $p(n)$ as a hypothesis (**inductive hypothesis**) and prove $p(n + 1)$.
- This is the delicate point, often subject to misunderstandings. “Assuming as a hypothesis” $p(n)$ does not mean assuming the thesis as a hypothesis. What must be proved is that:

“ for every $n \geq n_0$, if $p(n)$ is true then $p(n + 1)$ is also true ”

and not

“ if $p(n)$ is true for every n , then $p(n + 1)$ is also true ”

2 Proofs based on the principle of mathematical induction

2.1 Bernoulli's inequality

Remark 1: Bernoulli's inequality

For every integer $n \geq 0$, $x \in \mathbb{R}$, $x \geq -1$, we have:

$$(1 + x)^n \geq 1 + n x \quad (1)$$

Proof. By induction on n .

- **Base case of the induction**

Let $n = 0$. Then the statement becomes:

$$(1 + x)^0 \geq 1 + 0x \quad \text{i.e.} \quad 1 \geq 1$$

which is clearly true.

- **Inductive step**

Suppose it is true for n , and let us prove it for $(n + 1)$. By the inductive hypothesis, we have:

$$(1 + x)^n \geq 1 + nx$$

and moreover we have

$$(1 + x) \geq 0 \quad \text{since} \quad x \geq -1.$$

Then we can write:

$$\begin{aligned}(1 + x)^{n+1} &= (1 + x) \cdot (1 + x)^n \\ &\geq (1 + x) \cdot (1 + nx) \\ &= 1 + (n + 1)x + nx^2 \\ &\geq 1 + (n + 1)x\end{aligned}$$

where in the last inequality we used the fact that $nx^2 \geq 0$.

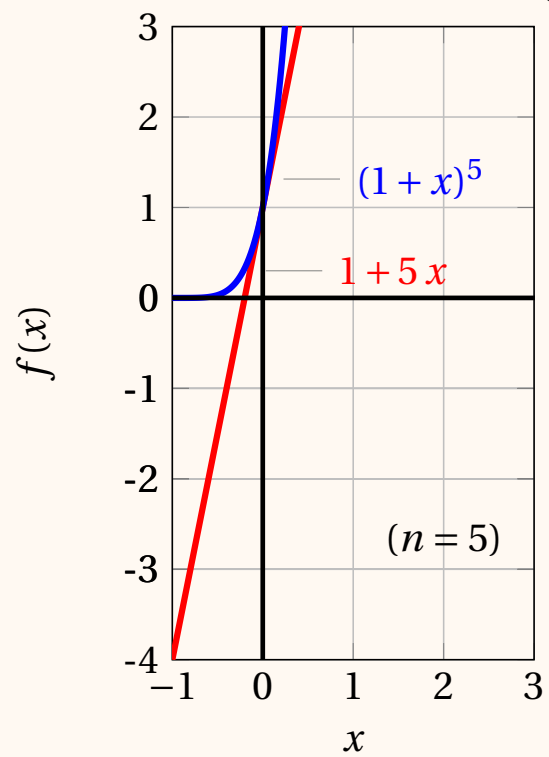
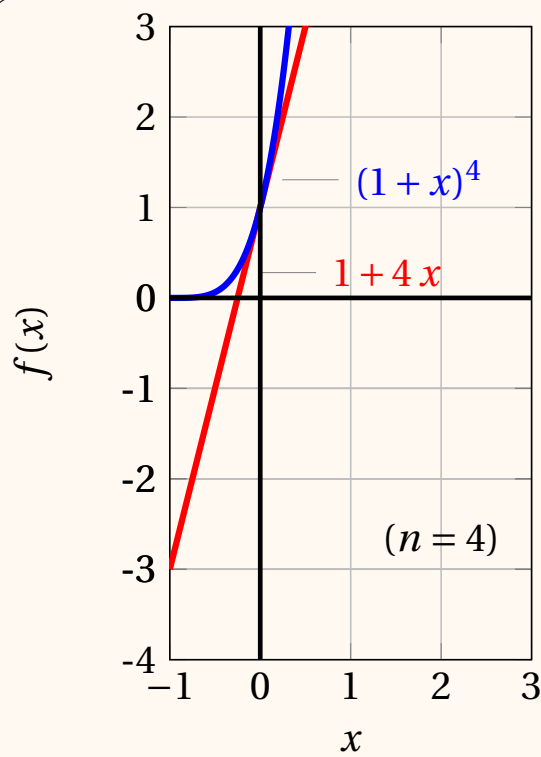
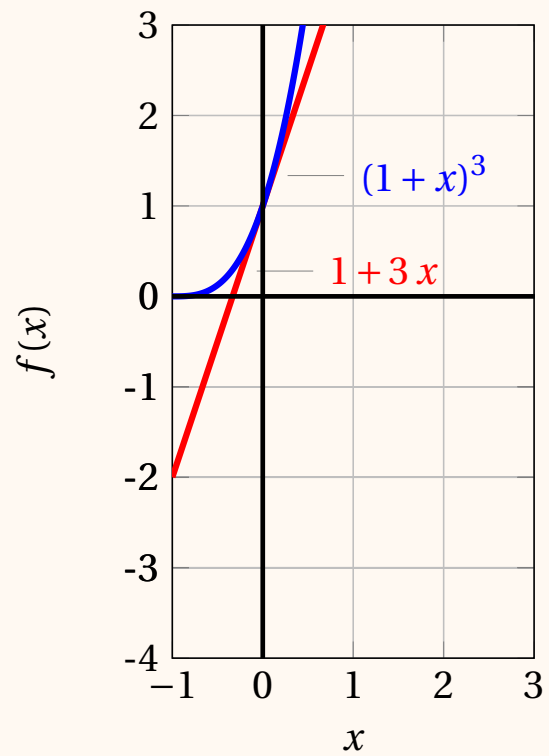
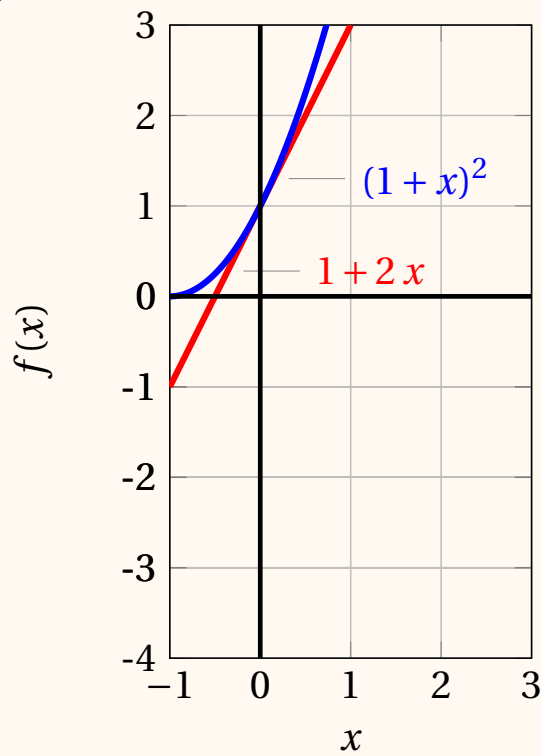
The chain of inequalities shows that, for $n + 1$, we have

$$(1 + x)^{n+1} \geq 1 + (n + 1)x$$

which is exactly the desired statement, for $n + 1$.

□

Example 1: Bernoulli's inequality



2.2 Some important summations

Remark 2: sum of the first n natural numbers

For every integer $n \geq 1$, we have:

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}$$

Proof. By induction on n .

- **Base case of the induction**

Let $n = 1$. Then the statement becomes:

$$\sum_{k=1}^1 k = \frac{1(1+1)}{2} \quad \text{i.e.} \quad 1 = 1$$

which is clearly true.

- **Inductive step**

Suppose it is true for n , and let us prove it for $(n+1)$. By the inductive hypothesis, we have:

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}.$$

Hence we can write:

$$\begin{aligned} \sum_{k=1}^{n+1} k &= \sum_{k=1}^n k + (n+1) \\ &= \frac{n(n+1)}{2} + (n+1) \\ &= \frac{n(n+1) + 2(n+1)}{2} \\ &= \frac{(n+1)(n+2)}{2} = \frac{(n+1)((n+1)+1)}{2} \end{aligned}$$

which is exactly the desired statement, for $n+1$.

□

2.3 Sum of the terms of the geometric progression

Remark 3: sum of the first n terms of the geometric progression ($a = 1$)

Given $q \in \mathbb{R}_+$, for every integer $n \geq 1$ we have:

$$\sum_{k=1}^n q^{k-1} = \begin{cases} \frac{q^n - 1}{q - 1} & \text{if } q \neq 1 \\ n & \text{otherwise} \end{cases} \quad (2)$$

Proof. By induction on n .

- **Base case of the induction**

Let $n = 1$. Then the statement becomes:

$$\sum_{k=1}^1 q^{k-1} = \frac{q^1 - 1}{q - 1} \quad \text{i.e. } 1 = 1$$

which is clearly true.

- **Inductive step**

Suppose it is true for n , and let us prove it for $(n + 1)$. By the inductive hypothesis, we have:

$$\sum_{k=1}^n q^{k-1} = \frac{q^n - 1}{q - 1}$$

Hence we can write

$$\begin{aligned} \sum_{k=1}^{n+1} q^{k-1} &= \sum_{k=1}^n q^{k-1} + q^n = \frac{q^n - 1}{q - 1} + q^n \\ &= \frac{q^n - 1 + q^{n+1} - q^n}{q - 1} = \frac{q^{n+1} - 1}{q - 1} \end{aligned}$$

which is exactly the desired statement, for $n + 1$.

□